

1. Design

I designed Entity A based on the textbook state diagram. Entity A receives the message from layer 5 with a queue. Then it sends a message from the queue to Entity B through layer 3. If a message times out, it resends the same message. The packets are created with checksum and seqnum. When it receives acknum from Entity B, it checks if acknum equals seqnum, and if it does then it will send a message from the queue.

I designed Entity B based on the textbook state diagram. Entity B receives the packet from Entity A and to check the checksum and seqnum are correct. If they are correct, it sends the message to layer 5, then increments the seqnum. Always send acknum to Entity A.

2. Testing

I added print statements and break points to determine what is going wrong. Based on the print statements, I can get ideas with which packets didn't run correctly. I also run it on the gradescope for testing.

In gradescope, I can also find which things are not working. Such as it will tell me the Entity B acknum is not correct. "108.3) Test run under 10000 messages with short interarrival time of 10. Corruption and loss both 10% (3/5)" This one does not work correctly, because with high values messages and it not running fast enough. "No. of Messages received at Entity B are not correct: Expected messages between 5499 and 9999. Got 5435 messages."

3. Output

```
C:\Users\Jiaku\Downloads\CS4480 Computer Networks>python3
rdtsim.py -d 100 -l 0.05 -c 0.05 -v 2
SIMULATION CONFIGURATION
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(-n) # layer5 msgs to be provided:    10
(-d) avg layer5 msg interarrival time: 100.0
(-z) transport protocol seqnum limit: 16
(-l) layer3 packet loss prob:         0.05
(-c) layer3 packet corruption prob:   0.05
```

(-s) simulation random seed: 1712114362972513500

===== SIMULATION BEGINS

TO_LAYER3: packet being lost

===== SIMULATION ENDS

SIMULATION SUMMARY

layer5 msgs provided to A: 10

elapsed time units: 1058.556977818595

layer3 packets sent by A: 11

layer3 packets sent by B: 10

layer3 packets lost: 1

layer3 packets corrupted: 0

layer5 msgs delivered by A: 0

layer5 msgs delivered by B: 9

layer5 msgs by B/elapsed time: 0.00850214035577623

A was supposed to send 10 layer 3 packets, and B was supposed to send 9 layer 3 packets. But 1 packet was lost, so both A and B need to send an extra layer 3 packet. That's why sent by A is 11 and sent by B is 10.